

EXAM 7 – SPRING 2015

2. (2.5 points)

Given the following paid claim information as of December 31, 2014:

Accident Year	Paid Claims (\$000)
2011	12,000
2012	11,250
2013	14,750
<u>2014</u>	<u>9,500</u>
Total	47,500

- The expected paid claim emergence pattern has been approximated by the following function where G is the cumulative proportion of ultimate claims paid and x represents the average time since accident occurrence in months.

$$G(x) = \frac{x}{x + 10}$$

- The expected incremental paid claim emergence follows an over-dispersed Poisson distribution with scaling factor $\sigma^2 = 25,000$.
- Parameter standard deviation for the total estimated unpaid claims is \$850,000.

a. (2 points)

Using a truncation point of 10 years, calculate the coefficient of variation of the total unpaid claims using the LDF method.

b. (0.5 point)

Identify the direction in which the coefficient of variation of the total unpaid claims estimate would change if the method used to calculate the unpaid claims estimate were changed from the LDF method to the Cape Cod method, and briefly explain the reason it would change in this direction.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 2

TOTAL POINT VALUE: 2.5

LEARNING OBJECTIVE: A2: Estimate parameters and unpaid claims using claims development models related to loss reserving methods such as chain ladder, Cape Cod, chain ladder plus calendar-year effects, Bornhuetter-Ferguson; A3: Calculate the moments and percentiles of unpaid claim distributions implied by the models.

SAMPLE ANSWERS

Part a: 2 points

Sample Answer 1

Accident Year	Paid Losses	Maturity	G(x)	Fitted LDF	Truncated Fitted LDF	Estimated Ultimate Losses	Estimated Unpaid Claims
		114	0.919	1.088	1.000		
2011	\$12,000	42	0.808	1.238	1.138	\$13,659	\$1,659
2012	\$11,250	30	0.750	1.333	1.226	\$13,790	\$2,540
2013	\$14,750	18	0.643	1.556	1.430	\$21,094	\$6,344
2014	\$9,500	6	0.375	2.667	2.452	\$23,290	\$13,790
Total	\$47,500					\$71,834	\$24,334

Note: All substantial values are in 000s

Maturity = Age of AY – 6

$G(x) = x / (x + 10)$ = given

Fitted LDF = $1 / G(x)$

Truncated LDF = $G(114) / G(x)$

Ultimate Loss = Paid Loss x Truncated LDF

Unpaid Loss = Ultimate Loss – Paid Loss

Scaling factor = $\sigma^2 = 25$ = given

Parameter SDev = 850 = given

Process variance = 608,343 = $\sigma^2 \times \text{Reserves} = 25 \times 24,334$

Process SDev = 780 = $\sqrt{\text{Process Variance}}$

Parameter variance = 722,500 = $(\text{Parameter SDev})^2 = (850)^2$

Total Variance = 1,330,843 = Process variance + Parameter variance

Total SDev = 1,154 = $\sqrt{\text{Total Variance}}$

Process CV = 4.74% = $\text{SDev} / \text{Reserves} = 1,154 / 24,334$

Sample Answer 2

$X_{\text{truncated}} = 12 \times 10 - 6 = 114$

$G(114) = 114 / (114 + 10) = 0.919$

Accident Year	Paid Losses	Maturity	G(x)	Estimated Ultimate Losses	Estimated Unpaid Claims
2011	\$12,000	42	0.808	\$13,649	\$1,649
2012	\$11,250	30	0.750	\$13,785	\$2,535
2013	\$14,750	18	0.643	\$21,081	\$6,331

EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

2014	\$9,500	6	0.375	\$23,281	\$13,781
Total	\$47,500			\$71,834	\$24,296

Note: All dollar values above are in 000s

Ultimate for 2011 = $\$12,000 / (0.808/0.919) = \$13,649$

Process variance = $6.074 \times 10^{11} = \sigma^2 \times \text{Reserves} = 25,000 \times \$24,296,000$

Total SDev = $\sigma = 1,153,213 = \sqrt{850,000^2 + \text{Process Variance}}$

CV = $0.0475 = 1,153,213 / 24,296,000$

Part b: 0.5 point

Sample Answer 1

The CV will be reduced. This is because we are relying on more information like premium or exposure, and this information allows us to make significantly better estimate of the reserve.

Sample Answer 2

CV should decrease because Cape Code uses more info (exposures) and uses a more stable LR for immature years instead of relying solely on possibly highly leveraged LDFs.

EXAMINERS' REPORT

Part a

- Candidates were expected to know how to estimate parameters and unpaid claims using claims development models related to Chain Ladder and Cape Cod loss reserving methods.
- Candidates generally knew how to set up and calculate the individual pieces required to calculate the coefficient of variation (CV).
- The most common error was keeping the total reserves in thousands and using the other inputs as whole dollars. Additional common errors included using the wrong truncation date, failing to truncate the LDFs, using ultimate losses in place of unpaid claim estimates, and applying the parameter standard deviation in place of parameter variance.

Part b

- Candidates were expected to know key assumptions of the models and how to test them, original Mack chain-ladder assumptions, relationship of variance assumptions to methods of calculating development factors, and how to test whether the methods work and how well the models fit.
- Candidates generally knew that the CV would be reduced by changing from the LDF method to the Cape Cod method. However, many were not able to give the correct explanation for this.
- Common errors included stating that Cape Cod has fewer parameters and therefore would have lower parameter variance and higher process variance. The original paper showed an example in which both the parameter and process variances were reduced (although the process variance was only slightly reduced). Clark did mention that it is possible for the Cape Cod method to have a "somewhat higher process variance". Some candidates wrote that the CV would increase or that the direction was uncertain.
- Other candidates argued that the reserves would be lower/higher for Cape Cod and therefore that would decrease/increase the CV.